

Supplementary material

Comfort-Constrained Operation of Multi-Mode HVAC in a Saudi Apartment Building under a Volume Tariff and Parameter Uncertainty

This supplement accompanies the main paper. Section S1 reports the passive zones of the whole building with and without the controller; Section S2 gives the perfect-foresight benchmark on every design day; Section S3 the building geometry and weather; Section S4 the formal definitions and running examples; Section S5 the details of the validation; Section S6 the full sensitivity analysis and the robustness tests of the pre-cooling result; Section S7 the uncertainty analysis; and Section S8 the calculation time. All values are generated from the same pipeline output as the main paper.

S1 Passive zones with and without the controller

The main paper imposes the comfort band on the conditioned rooms, and the passive zones exceed it on warm days. Table S1 shows how much of this excursion the controller removes without acting on them (32 baths of the whole building). With every unit off all year the baths average 39.3–41.3 °C in July and peak at 41.7–44.3 °C. With the controller holding their neighbours at the reference setpoint they average 24.7–24.9 °C in July, and their excursion above the band falls by 96%. They shed their heat to the conditioned neighbours through the walls and, above all, through the open doors (as the proposition on passive zones in the main paper predicts): with the interior doors closed the excursion is 4.3 times larger. A 1-ton unit in every bath would hold them in the band for 2.8–4.9% more energy, and colder neighbours (current practice) would cost 11.7–14.2% more.

Table S1: The baths of the SBC-compliant building (Case 3 of the main paper, the whole building; 32 baths) with and without the controller: July mean and annual peak temperature [°C], share of the year above the 24 °C ceiling [%], mean excursion above the ceiling per bath [K·h/yr], and the building’s HVAC energy relative to the controller of this study [%].

Case	Riyadh					Jeddah				
	July	Peak	> 24 °C	K·h	Energy	July	Peak	> 24 °C	K·h	Energy
No controller (all units off)	41.3	44.3	83	79,141	–	39.3	41.7	99	96,640	–
Controller, interior doors closed	27.7	31.9	68	15,164	+0.1	27.1	31.4	97	18,350	–0.3
Controller, reference setpoint (this study)	24.9	26.3	62	3,509	0	24.7	26.3	91	4,248	0
Controller at current practice (22 °C)	23.5	25.0	6	118	+14.2	23.3	24.9	4	52	+11.7
Controller and a 1-ton unit in every bath	23.6	24.0	0	0	+2.8	23.7	24.0	0	0	+4.9

S2 Perfect-foresight benchmark on every design day

Table S2 gives the perfect-foresight benchmark of the main paper on every design day solved; the main paper shows a selection of these rows, and the ranges it quotes are taken over all of them. The $\gamma = 0$ columns are rigorous upper bounds on what any schedule of the idealised model could save; the $\gamma = 0.02$ columns are achievable savings, those of the best valid schedule found by sequential linear programming from two starts (the program re-solved with the indoor factor of its first solution, and the program without pre-cooling), each schedule priced with its own indoor factor. Holding the ceiling is itself a valid schedule, so the achievable saving is at least 0, and a schedule that keeps the rooms within the band is valid down to 20 °C as well, so at 20 °C it is at least the saving at 22 °C of the same day. The bracket between the two columns holds for every periodic horizon solved, the design days and the 31-day July horizon (within the band). The search improves on holding the ceiling on 44 of the 48 design-day programs at $\gamma = 0.02$; on the others no valid schedule cheaper than the reference was found. Two checks support the linear model: linearising the door exchange about 0.2 or 1.0 K instead of 0.5 K moves the bound by at most 0.13 points, and the full nonlinear engine, run with ideal equipment holding the ceiling in its lowest mode, differs from the linear model’s reference by –3.1 to –0.9%; the schedule and its reference share the linear model, so this error bears on their level more than on the saving.

Table S2: Perfect-foresight benchmark (SBC-compliant building): saving [%] of the sensible cooling electricity that an ideal schedule achieves over holding the 24 °C ceiling with the same idealised units, with the rooms pre-cooled at most to the band floor (22 °C) or to the floor of the safe set (20 °C), with a COP that depends on the outdoor air only ($\gamma = 0$) or the COP of the engine ($\gamma = 0.02$, falling by 2%/K of indoor air below 27 °C). The $\gamma = 0$ columns are rigorous upper bounds on the saving of any schedule of the idealised model; the $\gamma = 0.02$ columns are achievable savings, those of the best valid schedule found (sequential linear programming, every schedule priced with its own indoor factor): at least the reference, 0, and at 20 °C at least the 22 °C schedule of the same day. Hold: electricity of the reference at $\gamma = 0.02$ [kWh/day]. Season: May–September mean days weighted by the reference energy of their month (Case 1); July, 31 days: the whole month as one horizon (Case 1, band floor only). Case 3: the bound only; the search at $\gamma = 0.02$ is not run. All design days.

Case	Day	Riyadh					Jeddah				
		Hold	$\gamma = 0.02$ (achievable)		$\gamma = 0$ (bound)		Hold	$\gamma = 0.02$ (achievable)		$\gamma = 0$ (bound)	
			22 °C	20 °C	22 °C	20 °C		22 °C	20 °C	22 °C	20 °C
1: apartment	May, mean day	16.8	0.13	0.13	0.79	1.47	13.7	0.06	0.06	0.59	1.04
	Jun, mean day	20.8	0.21	0.21	1.05	1.99	15.3	0.14	0.14	0.87	1.59
	Jul, mean day	22.3	0.28	0.28	1.26	2.41	16.8	0.10	0.10	0.78	1.43
	Aug, mean day	21.4	0.35	0.35	1.41	2.68	16.8	0.01	0.01	0.24	0.42
	Sep, mean day	17.1	0.13	0.13	0.82	1.52	15.0	0.01	0.01	0.34	0.56
	Jul, hottest day	25.4	0.46	0.47	1.50	2.86	19.3	0.54	0.54	1.79	3.34
	Season	–	0.23	0.23	1.07	2.07	–	0.06	0.06	0.58	1.03
	July, 31 days	22.3	0.33	–	1.31	–	16.9	0.23	–	0.96	–
2: floor	Jul, mean day	46.6	0.35	0.35	1.39	2.65	35.9	0.13	0.13	0.86	1.58
	Jul, hottest day	53.8	0.54	0.56	1.63	3.09	41.6	0.63	0.67	1.92	3.57
	Aug, mean day	44.9	0.43	0.43	1.56	2.95	36.2	0.00	0.00	0.27	0.47
Top floor	Jul, mean day	55.8	0.12	0.12	0.75	1.42	43.5	0.02	0.02	0.35	0.63
	Jul, hottest day	64.5	0.19	0.19	0.96	1.82	50.1	0.25	0.25	1.08	2.02
	Aug, mean day	53.7	0.16	0.16	0.82	1.56	43.8	0.00	0.00	0.07	0.12
3: building	Jul, mean day	204.8	–	–	0.94	1.78	157.3	–	–	0.50	0.91
	Jul, hottest day	235.6	–	–	1.16	2.21	181.1	–	–	1.34	2.50
	Aug, mean day	197.0	–	–	1.04	1.97	159.0	–	–	0.13	0.22

S3 Building geometry and weather

This section gives the room dimensions of the building and the monthly weather that the main paper summarises in its case-study section and its weather figure.

Room dimensions. Each apartment has a 5.00 × 5.00 m master bedroom reached through a 2.35 × 1.50 m lobby that also serves its 2.50 × 1.50 m bath; two 4.00 × 4.00 m bedrooms, Bed 2 on the north side beside the party wall and Bed 3 on the west side; a 2.00 × 2.00 m bath and a 2.00 × 2.00 m maid room along the party wall; a 5.00 × 5.00 m living room, open over its full width to a 2.20 × 5.00 m dining area on the party-wall side and joined by a sliding door to a 1.50 × 5.00 m west balcony enclosed by the exterior wall; a 4.00 × 4.00 m kitchen; a 6.90 × 5.00 m guest room; two further 2.00 × 2.00 m baths, one between the kitchen and the guest room and one beside the hall; and a corridor, hall and entrance that connect them (all clear dimensions). The two apartments of a floor mirror each other about a common stair (4.25 × 9.05 m) on the south side, on a 19.15 × 28.00 m footprint. Exterior walls are 0.30 m thick, the party and stair walls 0.25 m and partitions 0.15 m. Four floors of 3.40 m (3.00 m clear) form the building. Windows are 1.50 m high (sill 0.90 m, head 2.40 m), 0.80–2.00 m wide in the rooms and 4.00 m on the balcony, which gives a window-to-wall ratio of 17.3% for the building.

Weather. Table S3 gives the TMYx typical-year weather of Riyadh and Jeddah month by month. Riyadh’s monthly mean rises from 14.7 °C in January to 36.4 °C in July, with mean daily maxima above 43 °C in July and August and relative humidity near 12% in summer; Jeddah’s monthly mean never falls below 23.5 °C and its relative humidity stays between 54% and 70% all year.

S4 Formal definitions and running examples

This section gives the formal definitions on which the model and the theory of the main paper (Sections 5 and 6) rest, following the multi-mode formalism of Alur et al. (2012, 2013), Wojtczak (2013) and Mousa et al. (2016, 2017) and the coupled-zone model of Nghiem et al. (2011) (see the main paper’s reference list for these and the other works cited here), together with the continuations of the running example (Bed 2 of apartment 2A) and the heat budget of a passive zone. Symbols are those of the main paper; equation numbers in words refer to it.

Table S3: Measured climate driving the model (TMYx typical years): monthly mean, mean daily maximum and minimum dry-bulb temperature [$^{\circ}\text{C}$], mean relative humidity (RH) [%] and global horizontal irradiation (GHI) [kWh/m^2 per day].

	Riyadh					Jeddah				
	Mean	Max	Min	RH	GHI	Mean	Max	Min	RH	GHI
Jan	14.7	20.9	8.7	47	4.4	23.5	28.8	18.8	59	4.4
Feb	16.9	23.6	10.7	38	5.5	24.0	29.1	19.8	54	5.3
Mar	21.4	28.2	14.5	30	6.3	25.6	31.7	20.1	69	6.5
Apr	26.8	33.2	20.1	25	7.2	28.1	34.0	22.8	55	7.2
May	32.8	38.7	26.2	15	7.6	30.5	37.0	25.0	58	7.4
Jun	35.6	41.9	28.5	13	7.7	31.8	38.1	25.8	59	7.4
Jul	36.4	43.2	29.4	12	7.6	33.0	39.0	27.1	55	7.1
Aug	35.8	43.1	27.9	14	7.4	33.2	37.8	28.9	57	6.8
Sep	33.2	38.9	26.7	16	6.8	31.8	37.1	27.2	69	6.4
Oct	27.7	34.7	20.4	21	5.9	30.0	36.1	25.2	70	5.8
Nov	21.6	27.1	16.1	36	4.7	27.6	32.6	23.4	58	4.9
Dec	15.6	22.1	8.8	47	4.2	25.2	29.6	21.0	54	4.4

S4.1 Zones, units and the multi-mode system

Definition S4.1 (Coupled zone network). A building is a finite set of zones $\mathcal{Z} = \{1, \dots, N\}$ (rooms). The *conditioned* zones $\mathcal{Z}_c$ each have their own unit, and the *passive* zones $\mathcal{Z}_u$ (baths, maid room, lobby, corridor and hall) have none. Each zone i has a thermal capacitance C_i and a temperature $V_i(t)$. Two zones $i \neq j$ are *adjacent* if they share a partition wall or a floor slab; their coupling conductance $K_{ij} = K_{ji} \geq 0$ is the U-value of the shared element times its area, and $K_{ij} = 0$ otherwise. Zones with an exterior wall, window or roof (*edge zones*) exchange heat with the outdoors through a conductance $K_i^{\text{env}} > 0$; for interior zones K_i^{env} is the infiltration term alone. The temperature of each zone enters the dynamics of every adjacent zone: heat flows from zone j into zone i at the rate $K_{ij}(V_j - V_i)$ through shared walls and slabs and by buoyant air exchange through an open door between them, and leaks to the outdoors through the edge zones.

Definition S4.2 (Multi-directional multi-mode unit). Each conditioned zone $i \in \mathcal{Z}_c$ is served by a split heat-pump unit with four operating modes $M = \{0, 1, 2, 3\}$ (off, low, medium, high); a fixed-speed unit has only modes 0 and 3. The unit works in two directions: in mode m it moves heat at the rate $f_m \psi(T_a) Q_i$ out of the zone when cooling and $f_m Q_i^h$ into it when heating, with the capacity fractions $0 = f_0 < f_1 < f_2 < f_3 = 1$, and it draws $f_m \psi Q_i / \text{COP}_m$ or $f_m Q_i^h / \text{COP}_m$, with the cooling and heating COPs of the main paper (equation on the COP, Section 5). The thermostat decides whether a unit cools or heats. Passive zones have $f_m Q_i \equiv 0$.

Remark S4.1. The orderings $c_1 > c_2 > c_3$ and $h_1 > h_2 > h_3$ of the COP coefficients mean that a unit is most efficient at low output.

Definition S4.3 (Multi-mode system with costs). Following Mousa et al. (2017), the building is a multi-mode system $\mathcal{A} = (\mathbf{M}, N, A, \pi, V_{\min}, V_{\max}, V_0)$. Here $\mathbf{M} = M^{|\mathcal{Z}_c|}$ is the set of mode vectors, one mode per unit; $N = |\mathcal{Z}|$ is the number of continuous variables, the zone temperatures $V = (V_1, \dots, V_N)$; A gives the rate of change of V under each mode vector, which is the heat balance of the main paper (two-node heat balance, Section 5); $\pi_i(m)$ is the cost per unit time of unit i in mode m , given for the Saudi tariff in the main paper; the bounds $V_{\min}$ and $V_{\max}$ define the safe set $\mathcal{S} = [V_{\min}, V_{\max}]$ of the conditioned zones; and V_0 is the initial state.

Remark S4.2. Apart from the door exchange, which is nonlinear in the temperature difference, the rates are affine in the temperatures, so the building is close to a *linear-rate* multi-mode system in the sense of Wojtczak (2013); when the heat gain is held constant over a short interval (assumption A3) the rates are constant and the theory of Alur et al. (2012) applies.

S4.2 Timed actions, schedules, runs and costs

Definition S4.4 (Timed action and leap). A *timed action* is a pair $(m, t) \in M \times \mathbb{R}_{>0}$ of a mode m and a duration t : the unit stays in mode m for time t . A timed action is a *leap* when it moves the zone temperature from one bound of a band to the other, and a *drift* when $m = 0$ and the zone temperature floats.

Definition S4.5 (Schedule and infinite schedule). A *schedule* of horizon $t_{\max}$ is a finite sequence of timed actions $\sigma = \langle (m_1, t_1), \dots, (m_k, t_k) \rangle$ with $\sum_j t_j = t_{\max}$. An *infinite schedule* is either an infinite sequence $\langle (m_1, t_1), (m_2, t_2), \dots \rangle$ with $\sum_j t_j = \infty$, or a finite sequence whose last duration is $t_k = \infty$. Each unit follows its own schedule.

Definition S4.6 (Run, safe leap and safe schedule). The *run* of a schedule from the initial state $V^{(0)} = V_0$ is the sequence of states $V^{(0)}, V^{(1)}, V^{(2)}, \dots$ with $V^{(j)} = \Phi_{m_j}(V^{(j-1)}, t_j)$, the state reached by following the dynamics of mode m_j for time t_j from $V^{(j-1)}$; with constant rates, $V^{(j)} = V^{(j-1)} + t_j A(m_j)$. (The superscript indexes the run; the subscript i of V_i indexes zones. The labels $V_0, \dots, V_5$ of the schematic figure of the main paper are the states $V^{(0)}, \dots, V^{(5)}$ of one zone's run.) A timed action is a *safe leap* from a state if the trajectory stays in $\mathcal{S}$ for its whole duration. A schedule is *safe* if each of its timed actions is safe from the state its run has reached, and a safe schedule is *comfortable* if every conditioned zone also stays within the comfort band $[V_{\text{low}}, V_{\text{high}}] \subseteq \mathcal{S}$.

Definition S4.7 (Cost of a schedule). The cost of a finite schedule of unit i is $\pi(\sigma) = \sum_{j=1}^k \pi_i(m_j) t_j$. The cost of an infinite schedule is its limit-average cost

$$\pi_{\text{avg}}(\sigma) = \limsup_{k \rightarrow \infty} \frac{\sum_{j=1}^k \pi_i(m_j) t_j}{\sum_{j=1}^k t_j},$$

and, for a finite sequence ending in (m_k, ∞) , $\pi_{\text{avg}}(\sigma) = \pi_i(m_k)$. Since $\pi_i(m)$ is the energy price λ times the electrical power, $\pi_{\text{avg}} = \lambda J_{\infty}$, with J_{∞} the long-run mean electrical power of the unit.

Remark S4.3. The Saudi tariff has no charge per switch; Mousa et al. (2017) also attach a cost to each switch between modes, which is therefore zero here, and the energy a unit loses by cycling is counted in its electricity through the part-load factor.

Definition S4.8 (Leap cycle and optimal leap). Consider a conditioned zone in the cooling season with heat gain $L(V)$, the net heat entering it per unit time with its unit off. A *leap cycle* of amplitude $\Delta > 0$ between $V^{\top} - \Delta$ and $V^{\top}$ is a drift $(0, t_{\text{off}})$ from $V^{\top} - \Delta$ up to $V^{\top}$ followed by a cooling leap (m, t_{on}) back down; repeated for ever, it is an infinite schedule. Its *level* $\bar{V}$ is the time-mean temperature over the cycle, about $V^{\top} - \Delta/2$. Mode m *holds the zone* if $f_m \psi Q_i > L(\bar{V})$, that is, if it can bring the zone back. An *optimal leap* is a pair $(\bar{V}, m)$ whose mode holds the zone and whose cycle stays in the comfort band at the least cost π_{avg} . In the heating season the roles of drift and leap are reversed.

Definition S4.9 (Schedulability and utilisation). Following Alur et al. (2012, 2013), a zone is *safely schedulable* if it admits a safe infinite schedule. For states inside the safe set this holds if and only if there are dwell times whose rate vectors sum to zero over a period. For a single zone with drift rate $L/C_i > 0$ in mode 0 and rate $(L - f_m \psi Q_i)/C_i$ in mode m , this is the condition $f_m \psi Q_i > L$ of Definition S4.8. The fraction of time the unit must spend in mode m , $d_m = L/(f_m \psi Q_i)$, is its *duty* in that mode, and the zone is schedulable in mode m if and only if $d_m < 1$. Over a leap cycle the *utilisation* $U_i = L/Q_i$ is the mean fraction of the unit's full capacity in use.

Remark S4.4. Nghiem et al. (2012) give sufficient schedulability conditions for coupled zones that share a peak-power budget. Here each unit serves one zone and no budget is shared, so Definition S4.9 applies zone by zone.

Remark S4.5 (why the cycle returns to the switching point). The run of the schematic figure of the main paper returns to the thermostat's lower switching point rather than descending to V_{low} or V_{min} in every cycle, because a deeper cycle holds the room colder on average and so admits more heat: for the running example a cycle down to V_{low} costs 2.6% more and one down to V_{min} 7.7% more, although the unit switches far less often.

S4.3 Heat budget of a passive zone

Proposition S4.1 (Heat budget of a passive zone). Let $u \in \mathcal{Z}_u$ be a passive zone and let overbars denote means over a period long compared with its time constant (a month), over which its stored heat changes negligibly. Then the heat that enters it balances the heat that leaves it:

$$\bar{q}_u^{\text{env}} + \bar{G}_u + \bar{S}_u = \sum_j K_{uj} (\bar{V}_u - \bar{V}_j) + \bar{D}_u, \quad (1)$$

where the left-hand side is the mean heat that enters from the outdoors, the ground and the zone's own gains, the sum is the heat it loses through walls and slabs, and $\bar{D}_u$ is the mean heat it loses through its open doorways. Consequently: (i) in the cooling season a passive zone with any exposure or gain is warmer than the conditioned rooms that take its heat; (ii) its temperature does not depend on the capacity of any unit, only on the temperatures the units hold; (iii) the more of its heat it sheds directly to conditioned rooms, through walls or open doors, the closer it stays to their setpoint; and (iv) passive zones that touch one another raise one another's temperature.

Proof. Averaging the heat balance of u (where no unit acts) over the period gives the change of its stored heat divided by the period on the left, which vanishes when the stored heat changes negligibly, and the mean of the right-hand side on the right. The conductive terms are linear in the temperatures with constant coefficients, so their means are the same expressions evaluated at the mean temperatures; the door term is kept as its mean $\bar{D}_u$. Setting the sum to zero gives Eq. (1). In the cooling season the left-hand side is positive, so the right-hand side must be positive too; since both wall and door exchanges carry heat from warmer to cooler air, $\bar{V}_u$ must exceed the temperature of the rooms that receive its heat, which is (i). No unit capacity appears in Eq. (1), which is (ii). For a given inflow, a larger share of exchange with rooms held at the setpoint requires a smaller excess $\bar{V}_u - \bar{V}_j$, which is (iii). A passive neighbour is itself warmer than the setpoint by (i), so heat shed to it requires a higher $\bar{V}_u$, which is (iv). It also follows that $\bar{V}_u$ is a weighted mean of its neighbours' and the outdoor and ground temperatures, which the proof of the optimal-leap theorem in the main paper uses. $\square$

Running example (passive zones). In July in Riyadh ($\bar{T}_a = 36.4^\circ\text{C}$) the bath beside the kitchen of apartment 2A, with an exterior wall and a window, receives 127 W (80 W from the outdoors and 47 W of its own and solar gains). It sheds 22 W through its walls to conditioned rooms, 14 W to passive zones and 91 W through its open door, and averages 25.0°C . The least exposed bath, the shared bath, receives only 36 W and averages 24.4°C . By (i), each sits above the conditioned rooms that take its heat, and by (ii), a larger unit in a neighbouring room does not change that as long as the room is held at the same setpoint; Section S1 reports the passive zones of the whole building with and without the controller.

S4.4 Running example: further continuations

Coupled neighbours. Bed 2 is coupled to its neighbours with 138 W/K, about 7 times its envelope conductance, yet on the typical floor the zones on either side of the apartment's party and core walls differ by only 0.09 K on average over the year (Riyadh). Removing the coupling across those walls altogether changes the apartment's annual energy by 0.9%, and tripling it by -1.0% : a strong coupling carries little heat when no temperature difference drives it, as the remark on coupled neighbours in the main paper (Section 6) states.

Free-floating room and the indoor COP factor. With its unit off, Bed 2 would float towards 43°C in mean July conditions: its insulated wall loses little, so its own and solar gains alone would hold it far above the ambient. The mean powers of the main paper's running example are evaluated without the indoor COP factor g ($\gamma = 0$); with it, every mean power would be about 8% higher at 23.5°C and 11% at 22°C , which widens each difference.

Sources of saving. In July in Riyadh, Bed 2 gains 0.391 kW at 22°C and 0.363 kW at 23.5°C : each kelvin of warmth admits about 19 W less heat, and its mean electrical power falls from 83 to 77 W (lever a of the proposition on sources of saving). Holding the same level in mode 3 would cost 112 W (lever b). Pre-cooling the room admits the same 19 W per kelvin more, for as long as it lasts.

S5 Validation details

This section gives the details of the comparative tests of the main paper's validation: the BESTEST-style test on the single-zone cases of Standard 140 and the settings of the storey-box comparison with EnergyPlus.

BESTEST-style comparative test. The engine of the model was run on the single-zone cases 600 (lightweight) and 900 (heavyweight) of ANSI/ASHRAE Standard 140, with their geometry, constructions, 12 m^2 of south glazing, 200 W of internal gains, 0.5 ACH and the $20/27^\circ\text{C}$ setpoints, and with the units switched off (600FF and 900FF), on the Riyadh weather instead of the standard's Denver file, and the same cases were run in EnergyPlus 26.1 (Table S4). This is a comparative test between two programs on the same inputs rather than the standard's acceptance test, whose reference ranges apply only to Denver. Cases 640 and 940 add the standard's night heating setback (10°C from 23:00 to 07:00), which exercises the response of light and heavy construction to a setback. Annual cooling, which dominates in Riyadh, differs by -0.2% (case 600), $+4.3\%$ (case 900), -0.1% (case 640) and $+4.3\%$ (case 940), and the peak hourly cooling by -11.3% and -10.4% . Free-floating, the mean temperature differs by -1.2 and -1.3 K and the annual maximum by -5.5 and -4.5 K . The engine therefore reproduces EnergyPlus's annual loads closely and its peaks and extremes to within the spread usually found between programs: its exterior surfaces have a fixed film coefficient and no surface temperature of their own, and its structure is one node, so it damps the sharpest afternoon peaks. Heating is small in Riyadh, and its differences are a few hundred kWh.

Settings of the storey-box comparison. EnergyPlus runs with 10-minute time steps, its default conduction-transfer-function heat balance and surface convection algorithms, full exterior solar distribution and the same TMYx files as the model (Riyadh Air Base, WMO 404380, record 1958–2022; Jeddah King Abdulaziz International Airport, WMO 410240,

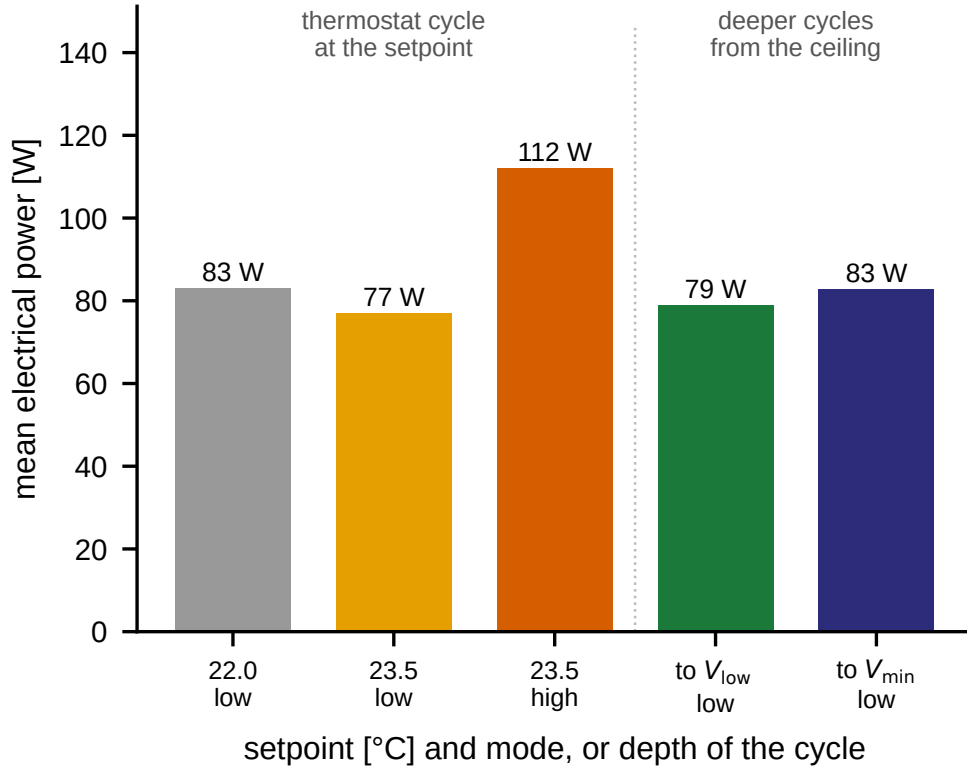


Figure S1: Leap cycles of the running example (moved from the main paper).

record 1981–2025). On the storey box the internal gains of the two programs are identical and their infiltration gains differ by 2.1–12.5%; the engine on the storey-box geometry finds an envelope effect of 53.6% in Riyadh and 53.0% in Jeddah, between those of EnergyPlus and of the room-resolved model.

S6 Full sensitivity analysis and pre-cooling robustness

This section gives the full one-at-a-time sensitivity analysis of the main paper (Table S5; the main paper shows the decisive rows), the diagnostics behind it (sizing margin, controller resolution and humidity), the seasonal operation of the building, every test instance, and the tests of the pre-cooling result where it is most likely to fail. All runs are of the whole building (Case 3); the sensitivity variants and the robustness tests are of the SBC-compliant building.

Sizing margin. Units sized by load are sized at a multiple of each room’s peak hourly load on the sampled days; of the margins tried in turn (1.15, 1.5, 2, 2.5), the first at which every room stays within the safe set is 2.5 in both cities. A finer

Table S4: BESTEST-style comparative test (Standard 140 cases on the Riyadh weather; differences computed from unrounded values): annual heating and cooling [MWh], peak hourly heating and cooling [kW], and the free-floating minimum, mean and maximum temperature [°C]. Cases 640/940: 600/900 with the night heating setback. E+: EnergyPlus 26.1; this: the engine of the model.

	Case 600		Case 900		Case 640		Case 940	
	E+	this	E+	this	E+	this	E+	this
Heating [MWh]	0.40	0.62	0.00	0.01	0.16	0.28	0.00	0.00
Cooling [MWh]	11.12	11.10	9.62	10.04	11.05	11.04	9.62	10.03
Peak heating [kW]	1.38	1.63	0.26	0.57	3.30	3.77	0.25	1.05
Peak cooling [kW]	6.57	5.83	4.02	3.60	6.57	5.83	4.02	3.60
Case 600FF			Case 900FF					
Minimum [°C]	12.0	9.6	20.2	19.0				
Mean [°C]	39.1	37.9	39.2	37.9				
Maximum [°C]	71.1	65.6	51.6	47.1				

Table S5: Sensitivity analysis (SBC-compliant building, Case 3): HVAC energy intensity [kWh/m²yr], setpoint lever, saving of the best thermostat pre-cool (0.5–2 K in either window), and the change in cost of a 2 K pre-peak pre-cool. C_d : cycling degradation coefficient; sized by load: at 2.5 times the peak hourly load, the first tested margin that keeps every room within the safe set, unless stated; –: inadmissible, the reference policy lets a room pass the ceiling, so the run is excluded and nothing but its excursion (in the text) is reported. Door schedule: bath doors closed, bedroom and guest-room doors closed 23:00–07:00.

Variant	Riyadh				Jeddah			
	Int.	Setp.	Pre-cool	2 K	Int.	Setp.	Pre-cool	2 K
Reference (base case)	25.1	12.4%	0.00%	+3.4%	37.7	10.5%	0.00%	+4.3%
Interior doors closed	25.1	12.2%	0.00%	+3.1%	37.6	10.5%	0.00%	+3.9%
Internal gains –30%	23.0	11.4%	0.00%	+3.5%	35.0	11.0%	0.00%	+4.3%
Internal gains +30%	27.4	12.7%	0.00%	+3.2%	40.4	10.1%	0.00%	+4.3%
Flat part-load COP	31.5	12.6%	0.00%	+2.1%	45.0	11.0%	0.00%	+3.2%
Ground at constant mean temperature	26.0	17.0%	0.00%	+4.0%	42.2	12.5%	0.00%	+4.6%
One node per zone	25.5	11.2%	0.00%	+3.3%	38.0	10.4%	0.00%	+6.7%
One node, COP independent of indoor air	23.8	8.2%	0.08%	+1.8%	36.1	8.0%	0.00%	+5.0%
Air node 2% of C	24.7	12.5%	0.00%	+4.6%	37.2	10.7%	0.00%	+5.4%
Air node 15% of C	25.6	12.2%	0.00%	+2.7%	38.6	10.4%	0.00%	+3.7%
Air node 30% of C	25.5	12.3%	0.00%	+2.8%	38.5	10.5%	0.00%	+4.2%
Air–structure coupling $\times 0.5$	24.5	12.0%	0.00%	+3.3%	36.8	10.4%	0.00%	+3.8%
Air–structure coupling $\times 2$	25.5	12.6%	0.00%	+3.0%	38.3	10.5%	0.00%	+4.6%
Latent COP –30% off / low mode	25.1	12.4%	0.00%	+3.4%	41.6	9.5%	0.00%	+3.8%
Thermostat differential 0.1 K	26.4	11.4%	0.00%	+4.3%	38.8	10.5%	0.00%	+5.1%
Escalation at 0.2 / 0.6 K	26.0	12.6%	0.00%	+3.1%	38.5	10.8%	0.00%	+4.4%
Escalation at 0.8 / 2.0 K	25.0	12.3%	0.00%	+2.3%	37.7	10.4%	0.00%	+3.2%
Inverter units in low mode only	25.0	12.3%	0.00%	+2.0%	37.7	10.4%	0.00%	+2.8%
No full-capacity response	25.1	12.4%	0.00%	+3.4%	37.7	10.5%	0.00%	+4.3%
COP independent of indoor air	23.6	9.5%	0.00%	+2.2%	36.0	8.1%	0.00%	+3.0%
Indoor COP factor 3%/K	25.9	14.0%	0.00%	+4.1%	38.6	11.8%	0.00%	+5.1%
No cycling loss ($C_d = 0$)	21.3	12.5%	0.00%	+4.0%	33.2	10.1%	0.00%	+4.8%
$C_d = 0.25$ for every unit	26.4	12.4%	0.00%	+3.2%	39.3	10.5%	0.00%	+4.1%
3-ton units as inverter units	21.3	12.4%	0.00%	+4.5%	33.4	10.0%	0.00%	+5.2%
No standby power	24.3	12.8%	0.00%	+3.5%	36.9	10.7%	0.00%	+4.4%
No capacity derating	25.1	12.4%	0.00%	+3.3%	37.8	10.6%	0.00%	+4.1%
Indoor fan on while compressor off	28.0	10.9%	0.00%	+3.3%	43.2	8.8%	0.00%	+4.1%
Sized by load, all inverter	20.9	11.7%	0.00%	+2.9%	33.5	9.6%	0.00%	+3.0%
Sized by load, fixed-speed kept	24.5	11.9%	0.00%	+2.6%	37.6	10.1%	0.00%	+2.9%
Sized by load, 15% margin	–	–	–	–	–	–	–	–
Wall U-value +20% (thermal bridges)	25.6	12.3%	0.00%	+3.4%	38.2	10.6%	0.00%	+4.3%
Solar absorptance 0.5	24.3	12.3%	0.00%	+3.6%	36.8	10.6%	0.00%	+4.4%
Solar absorptance 0.9	25.9	12.5%	0.00%	+3.3%	38.6	10.4%	0.00%	+4.3%
Infiltration 0.2 ACH	22.5	12.9%	0.00%	+3.2%	30.4	10.7%	0.00%	+4.5%
Infiltration 0.8 ACH	31.5	10.0%	0.00%	+3.4%	52.4	10.2%	0.00%	+3.8%
Door schedule	24.8	12.3%	0.00%	+3.6%	37.4	10.4%	0.00%	+4.5%
Daily swing $\times 1.3$	26.0	11.8%	0.00%	+3.2%	38.2	10.4%	0.00%	+4.6%
Heat wave +3 K, swing $\times 1.3$	33.4	12.2%	0.00%	+3.1%	46.7	9.7%	0.00%	+4.1%

sweep of the margin (1.15, 1.25, 1.5, 1.75, 2, 2.5, 3) shows what it buys. At the margins that keep every room within the safe set (2.5, 3) the units save 15.5–16.5% in Riyadh and 10.6–11.2% in Jeddah against the catalogue units; the smaller margins are inadmissible, and only their excursion is reported: it falls from 69 K·h in Riyadh and 18 K·h in Jeddah at 1.15 to zero from 2.5 on, so the smallest margin that keeps every room within the safe set lies between 2 and 2.5. Every unit is at the 0.75-ton minimum size at 1.15, so the building needs 45 tons at 1.15 and 63–66 at 3: in capacity the margins differ far less than their names suggest. The excursions are small, the warmest room peaking at 24.15 °C, and mostly in the guest rooms (98–100% of the excursion at the tightest margin). Starting each unit at full capacity leaves them unchanged, so they do not arise at start-up: they arise while a running unit escalates from a lower mode, with the switch-on point at the ceiling itself (see the main paper’s sensitivity analysis for what this implies for the margin). The change of compressor, which the 3-ton variant measures on its own, accounts for most of the difference between the units sized by load as inverter units (16.5% in Riyadh and 11.2% in Jeddah against the catalogue units) and with the fixed-speed zones keeping fixed-speed units (2.1% in Riyadh and 0.2% in Jeddah), the two effects not being additive.

Controller resolution. The units start on average 2.0 (Riyadh) and 2.6 (Jeddah) times per hour of the year, a rate set by the 3-minute time step rather than by a measured cycle; without the 3-minute restart delay the bill changes by less than 0.01%. The twelve fixed-speed 3-ton units run 3–6% of the time and as inverter units would use 39% less; the four stair units, a feature of the building studied rather than of every Saudi apartment block, take 8.9–9.1% of the units’ electricity and are billed to the common meter. The robustness tests below show how far a 1-minute step, minimum on- and off-times and a cycling loss charged per start move the results.

Humidity. The main model prices the latent load at a fixed latent COP and assumes the indoor target is met. A diagnostic humidity balance per zone, with infiltration and the internal moisture of five occupants per dwelling entering and the coil removing moisture at a sensible heat ratio while the unit runs, tests that assumption. In Riyadh the target holds: the conditioned rooms exceed 60% relative humidity in 0.8% of the cooling-season hours. In Jeddah the sensible runtime of the oversized units is too short: at a sensible heat ratio of 0.75 the rooms exceed 60% in 41% of the hours (21% at 0.65), and the coil removes 82% of the latent load, so holding the target needs dedicated dehumidification, whose electricity the latent term prices. Internal latent gains would add 0.2% (Riyadh) and 3.2% (Jeddah) to the HVAC energy at every schedule, leaving every absolute saving unchanged. A 30% lower latent COP whenever the units are off or in the lowest mode raises Jeddah’s latent share to 33% and leaves the scheduling result unchanged. Run for the levers themselves (current practice, the reference setpoint and 2 K of pre-peak or night pre-cooling), the diagnostic hardly moves: at a sensible heat ratio of 0.75 the rooms exceed 60% in 0.7–0.9% of the cooling-season hours in Riyadh and 41.1–44.6% in Jeddah. At that ratio the higher setpoint lowers the Jeddah share slightly, since warmer air holds the same moisture at a lower relative humidity, while at 0.65 it raises it by 1.3 points; pre-peak pre-cooling raises it at both ratios, the coil then removing 76% of the latent load instead of 82%, so the setpoint lever costs little humidity comfort and pre-cooling gains none.

Seasonal operation. The model runs through the whole year with units that cool and heat (Table S6). In Riyadh summer accounts for 61% of apartment 2A’s HVAC energy and winter for 4%; Jeddah spreads its load more evenly (47% and 13%). Winter is a heating season only in Riyadh, and a substantial one only for the pre-code building, which uses 11,116 kWh of heating a year; the SBC-compliant building needs 1,367 kWh, because its internal gains and its neighbours keep most rooms above the heating setpoint. At the reference policy no conditioned room of the SBC-compliant building leaves the safe set in either season (cold excursion 0 and 0 K·h). Inverter units use the medium and high modes on 4.5–5.2% of the hours; the time shares of Table S6 include the fixed-speed units, which run only at full capacity. The latent load, computed from the measured humidity, is 0.4% of the HVAC energy in Riyadh and 26% in Jeddah.

Table S6: Summer and winter operation of the whole building at the reference policy: annual cooling, heating and latent electricity [kWh], and the share of time [%] the conditioned rooms spend off and in the low, medium and high modes (cooling and heating together; fixed-speed units count as high) in summer (Jun–Sep) and winter (Dec–Feb). The remainder of the HVAC energy is the units’ standby electricity, 1,577 kWh a year in the SBC-compliant building. Last column: summer / winter shares of apartment 2A’s HVAC energy.

City	Envelope	Cooling	Heating	Latent	Summer modes	Winter modes	Apt. 2A summer / winter
					off/low/med/high	off/low/med/high	
Riyadh	SBC	47,383	1,367	185	59/33/1/7	97/2/0/1	61.0% / 3.9%
	pre-code	103,038	11,116	185	40/32/15/13	81/14/1/5	–
Jeddah	SBC	54,784	0	19,551	63/31/0/6	82/14/0/4	46.8% / 12.6%
	pre-code	116,870	0	19,551	44/32/13/12	75/18/2/6	–

Test instances. Table S7 lists every test instance; the main paper reports the eight instances with the catalogue units in one row.

Table S7: Every normal and hard test instance (Case 3, the whole building). Units: catalogue (sized by coverage) or sized to the building's own peak hourly load with a 15% margin. Peak load ratio: largest hourly sensible load of any unit over its rated capacity; excursion of the reference policy above 24 °C and below 20 °C; best policy: the cheapest admissible thermostat policy, night/pre-peak depth (none: no pre-cooling; –: no lattice setting at the reference setpoints is admissible; in brackets, the highest admissible cooling setpoint without pre-cooling, where one exists); admissible policies: number of the 49 lattice policies that keep every conditioned room within the safe set on the sampled days (the others are excluded). No unit reaches a load ratio of 0.85.

Envelope	City	Units	Weather	Class	Peak load ratio	Exc. above/below [K·h]	Best policy	Admissible policies
SBC	Riyadh	catalogue	TMYx	normal	0.19	0/0	none	49 of 49
SBC	Jeddah	catalogue	TMYx	normal	0.19	0/0	none	49 of 49
SBC	Riyadh	catalogue	TMYx +3 K	normal	0.26	0/0	none	49 of 49
SBC	Jeddah	catalogue	TMYx +3 K	normal	0.26	0/0	none	49 of 49
Pre-code	Riyadh	catalogue	TMYx	normal	0.55	0/0	none	42 of 49
Pre-code	Jeddah	catalogue	TMYx	normal	0.53	0/0	none	36 of 49
Pre-code	Riyadh	catalogue	TMYx +3 K	normal	0.59	0/0	none	30 of 49
Pre-code	Jeddah	catalogue	TMYx +3 K	normal	0.62	0/0	none	33 of 49
SBC	Riyadh	by load	TMYx	hard	0.58	69/0	– (23 °C)	0 of 49
SBC	Jeddah	by load	TMYx	hard	0.57	18/0	– (23 °C)	0 of 49
Pre-code	Riyadh	by load	TMYx	hard	0.79	81/0	– (23 °C)	0 of 49
Pre-code	Jeddah	by load	TMYx	hard	0.75	29/0	– (23 °C)	0 of 49

Robustness of the pre-cooling result. Table S8 collects the tests of the thermostat pre-cooling result where it is most likely to fail (SBC-compliant building, Case 3). None of them saves, except the favourable combinations, which need every favourable choice at once and save at most 0.77%. The loss per start is the unit's full power for 1.5 min at each start of a fixed-speed unit and 0.82 min of an inverter unit, calibrated to the part-load factor at the cyclic rating point; the controller variants move the absolute bill by 2–32%, the loss per start most, but not the sign of pre-cooling. With minimum on- and off-times the variant is inadmissible in Riyadh: its reference breaks the ceiling in the heating months (December to February), when the oversized fixed-speed units, held on for 3 minutes while heating, overshoot it (636 K·h a year). Scaled to EnergyPlus's extra heat, a 2 K night pre-cool still costs 3.2% in Riyadh and 3.3% in Jeddah. With the favourable choices combined, the best ramped setback changes the bill by –0.55% (Riyadh) and +0.29% (Jeddah); with the 0.08% of the one-node variant whose COP ignores the room temperature (Table S5), these rows hold the only savings found for thermostat pre-cooling. No policy cools a room of the SBC-compliant building below the 20 °C floor; in the pre-code building, on the sampled days, 7 (Riyadh) and 13 (Jeddah) of the 49 policies are inadmissible and are excluded: 7 in Jeddah let rooms drift below the floor on cool nights, by at most 0.2 K·h a year and to 19.9 °C, and the remaining 7 in Riyadh and 6 in Jeddah pass the ceiling. The cheapest admissible policy remains no pre-cooling.

Table S8: Robustness of the thermostat pre-cooling result (SBC-compliant building, Case 3, volume tariff). Best saving: saving of the cheapest admissible policy against no pre-cooling [%] (positive: saves; 0: no pre-cooling is best; negative: every depth run costs, for the rows that do not include no pre-cooling among their options; –: not run, or inadmissible because the reference policy leaves the safe set). 2 K pre-peak: change in the annual bill of a 2 K pre-peak pre-cool [%] (positive: costs). Favourable choices: one node per zone, COP independent of the indoor air, inverter units in their lowest mode, no cycling loss. Dawn setback: all 365 days; the other rows: sampled year.

Test	Riyadh		Jeddah	
	Best saving	2 K pre-peak	Best saving	2 K pre-peak
Base model, lattice of 49 policies	+0.00	+3.40	+0.00	+4.30
Inverter units in their lowest mode	+0.00	+1.95	+0.00	+2.84
Ramped pre-peak setback (0.5 K/h)	–0.25	+2.26	–0.34	+3.04
Extra heat scaled to EnergyPlus	–	+2.94	–	+3.82
Dawn setback, 2 K, all 365 days	–0.99	–	–1.80	–
Top floor only (15 units)	+0.00	+0.82	+0.00	+1.14
Rooms with west windows only (12 units)	+0.00	+1.02	+0.00	+1.29
Cycling loss charged per start	+0.00	+2.45	+0.00	+2.82
1-minute step	+0.00	+2.11	+0.00	+2.85
1-minute step, 3-min minimum on/off, loss per start	–	–	+0.00	+2.45
Favourable choices combined	+0.77	–0.21	+0.03	+3.07
Favourable choices, top floor only	+0.21	–0.20	+0.00	+0.50

S7 Uncertainty analysis

Seventeen assumed inputs of the SBC-compliant building (Case 3, sampled year) were varied jointly over the uniform ranges of Table S9 by a Latin hypercube of 128 samples per city (fixed seed). This section gives the input ranges and the partial rank correlations; the per-sample inputs and results are not reproduced here but are in the repository (results/parts/uncertainty_Riyadh.json and uncertainty_Jeddah.json). For each sample the reference, current-practice and 0.5–2 K pre-cooling policies (night or pre-peak), the pre-code building and the 3-ton units as inverter units were simulated. The weather inputs scale the daily swing of the air temperature about each day’s mean and shift it uniformly (the heating months follow the shifted weather). Table S10 gives the partial rank correlation coefficient (PRCC) of every input with every result.

Table S9: Inputs of the uncertainty analysis: base value and uniform range.

Input	Base	Low	High
indoor COP slope γ	0.02	0	0.03
outdoor COP slope	0.02	0.01	0.03
COP ratio c_1/c_3	1.45	1.3	1.6
COP ratio c_2/c_3	1.24	1.1	1.3
inverter C_d	0.15	0.05	0.25
fixed-speed C_d	0.25	0.15	0.35
escalation thresholds (scale)	1	0.5	2
air-node share of capacitance	0.05	0.02	0.15
latent COP	3.4	2.5	4
infiltration (ACH)	0.4	0.2	0.8
internal gains (scale)	1	0.7	1.3
wall U (thermal bridges)	1	1	1.2
opaque solar absorptance	0.7	0.5	0.9
door exchange (scale)	1	0.5	1.5
capacity derating slope	0.01	0.005	0.02
daily temperature swing (scale)	1	0.8	1.3
air temperature shift [K]	0	−1	2

Table S10: Partial rank correlation coefficients of the inputs with the results (Riyadh / Jeddah). Columns: setpoint lever, envelope lever, inverter lever, best admissible pre-cool (omitted when no sample saves), cost of a 2 K pre-peak and of a 2 K night pre-cool.

Input	Setp.	Env.	Inv.	Pre-peak	Night
indoor COP slope γ	+0.93/+0.98	+0.04/+0.10	−0.17/−0.09	+0.69/+0.65	+0.76/+0.81
outdoor COP slope	−0.19/−0.10	+0.23/+0.39	+0.01/+0.09	−0.14/+0.34	−0.78/−0.80
COP ratio c_1/c_3	−0.32/−0.28	+0.43/+0.33	+0.90/+0.89	+0.35/+0.34	+0.12/+0.16
COP ratio c_2/c_3	−0.12/−0.09	+0.05/−0.10	+0.13/+0.17	+0.10/+0.03	+0.07/+0.07
inverter C_d	−0.19/+0.16	−0.36/−0.04	−0.89/−0.89	−0.37/−0.38	−0.15/−0.16
fixed-speed C_d	+0.04/+0.22	+0.04/+0.03	+0.94/+0.95	−0.08/−0.14	−0.13/−0.16
escalation thresholds (scale)	−0.22/−0.27	−0.55/−0.27	+0.32/+0.33	−0.36/−0.34	−0.24/−0.19
air-node share of capacitance	−0.36/−0.41	−0.80/−0.80	+0.31/+0.05	−0.44/−0.46	+0.08/−0.04
latent COP	−0.10/+0.84	+0.12/+0.77	+0.14/+0.64	−0.06/+0.16	+0.04/+0.33
infiltration (ACH)	−0.85/−0.57	−0.88/−0.96	−0.21/−0.82	+0.34/−0.27	+0.55/+0.15
internal gains (scale)	+0.48/−0.80	−0.90/−0.77	−0.15/+0.01	−0.26/−0.24	−0.29/−0.37
wall U (thermal bridges)	+0.00/+0.06	+0.35/+0.00	−0.07/+0.07	−0.04/−0.03	+0.02/−0.03
opaque solar absorptance	−0.07/−0.43	+0.85/+0.87	+0.11/+0.18	+0.05/+0.09	−0.00/+0.03
door exchange (scale)	−0.03/−0.11	−0.14/+0.11	−0.15/−0.29	+0.02/+0.03	+0.05/+0.01
capacity derating slope	+0.06/+0.04	−0.31/+0.15	−0.05/−0.03	+0.09/+0.12	+0.15/+0.22
daily temperature swing (scale)	−0.43/−0.20	+0.05/−0.03	+0.07/+0.13	−0.15/+0.12	−0.50/−0.55
air temperature shift [K]	+0.73/−0.78	−0.52/+0.78	+0.12/+0.36	+0.02/−0.28	+0.21/−0.42

S8 Calculation time

Table S11 reports the measured calculation times. One annual evaluation of the whole building (132 zones with two nodes each, 120 open doors, 28,800 integration steps) takes 13.4 s for one policy and 52.5 s for the 49 policies of the lattice together, i.e. 1.07 s per policy. The simulation time grows much more slowly than the number of zones, because at these sizes the fixed cost of each step dominates. The optimal-control program for one day has 5,088 variables and 3,072 equality constraints in Case 1 and 10,656 and 6,336 in Case 2, and takes 1.3 and 18.4 s for the bound ($\gamma = 0$, one solve), and the search for the indoor COP factor 21.2 and 284.3 s (19 and 19 solves; the top floor under the roof, the size of Case 2, 295.1 s; the bound of the whole building, with Clarabel, 233.8 s), against 18.9 and 24.2 s for a year of all 49 thermostat policies; the program’s time grows faster with its size, about 13-fold for twice the zones. In the building each unit needs only its own temperature and the clock: selecting the modes of all 60 units at one control step takes 8.4 μ s; the call overhead of one

vectorised step dominates at all three sizes, so the time does not grow with the number of units (the smaller cases are within timing noise), and a single unit’s thermostat logic is trivial for ordinary hardware.

Table S11: Calculation time (two-core virtual machine): one annual evaluation of a thermostat policy (sampled year, 3-minute steps), alone and for all 49 policies together; the optimal-control linear program for one mean July day (15-minute steps; the bound, $\gamma = 0$, and the search, sequential linear programming for the indoor COP factor; Case 3: bound only, with Clarabel); and the controller at one control step.

Case	Zones	Units	Doors	1 policy [s]	49 policies [s]	Bound/search [s]	Ctrl. [μ s]
1	16	7	15	11.2	18.9	1.3/21.2	7.2
2	33	15	30	10.0	24.2	18.4/284.3	7.1
3	132	60	120	13.4	52.5	233.8/–	8.4